

Your heat pump proposal

Thanks for choosing us for your heat pump installation!

A heat pump will give you reliable, even and comfortable heating, will use much less energy than a boiler, and will greatly reduce your carbon footprint.

Please read the enclosed quote carefully. It covers in detail the components we will be installing, describes how the system works, shows how the system will meet the heating requirements of your property, and lists the costs for the work.

We're always happy to help answer questions, so please get in touch if there is anything you aren't sure about.

Tej Patel

This proposal is for:

John Test

103 High View Watford
Hertfordshire
WD18 6JP

Prepared

9 April 2026

Valid for

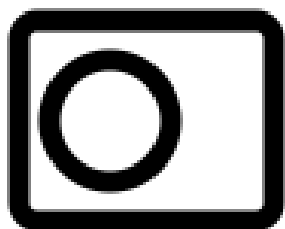
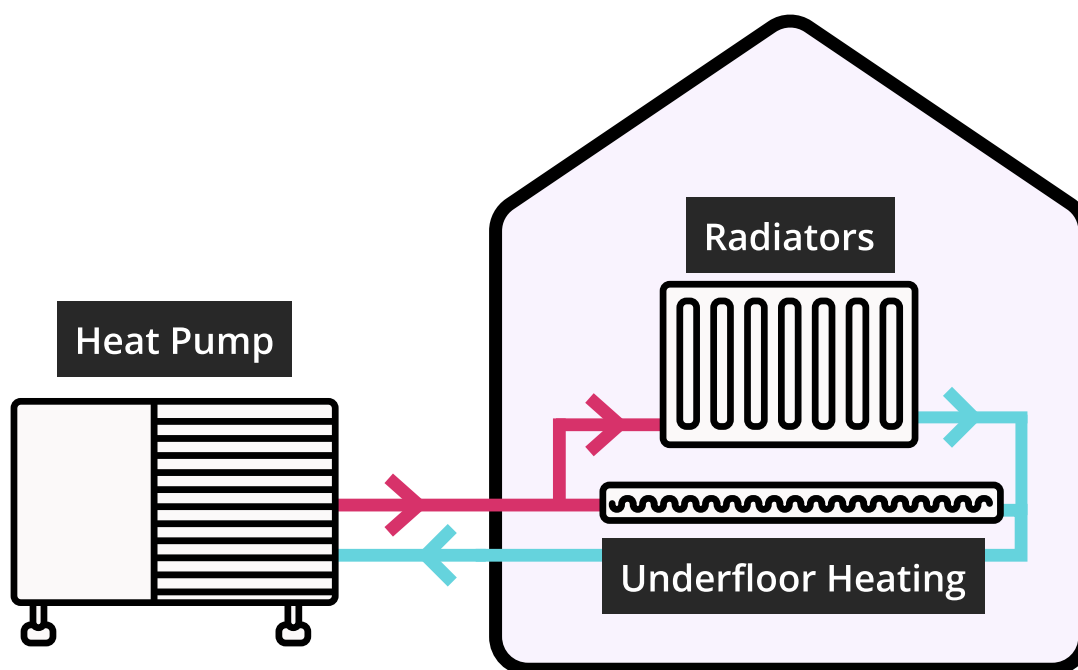
30 days

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Your system

The system we propose includes a heat pump (Vaillant aroTHERM plus 3.5), and all the other components necessary to plumb the heat pump into your heating system. We have also specified underfloor heating which will help your new system run more efficiently.



Vaillant aroTHERM plus 3.5

With a SCOP of up to 4.41, the aroTHERM plus is extremely energy efficient. The 3.5 is the smallest heat pump in the range.

ENA Registration Number: HP_1127

Sound Power Level: 54.0 dB

Proposed flow temperature: 50°C

*at outside design temperature of -2°C

Heating SCOP at 50°C: 3.37

Nominal Output: 3.50 kW

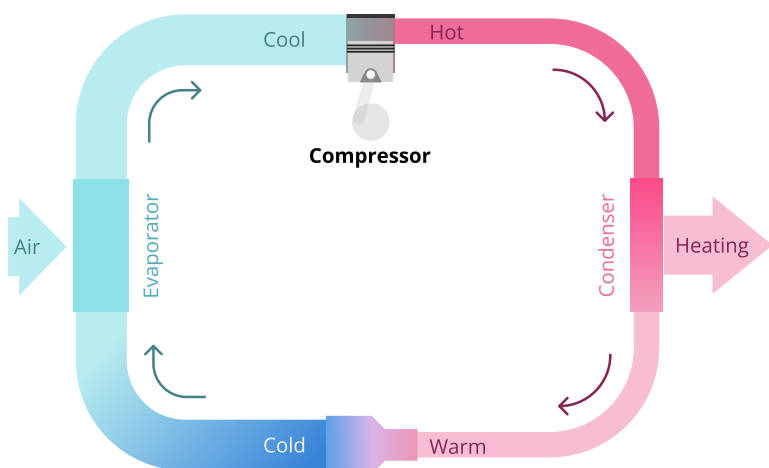
Actual Output at 50°C*: 4.30 kW

Heat Pumps Explained

A heat pump uses electricity to move heat from the outside of a building to the inside. It works on the same principle as a fridge.

In a heat pump, a compressor is used to pressurise a refrigerant, making it hotter - in the same way that a bicycle pump generates heat when it compresses air. The hot refrigerant is passed through a heat exchanger that draws off the heat for use in your house.

The refrigerant (now much cooler) is then passed through an evaporator nozzle, which allows it to expand. As the pressure decreases the refrigerant cools even further - in the same way that aerosols get cold when compressed air is released from a can.



The beauty of heat pumps is that a relatively small electrically-driven compressor can move large amounts of heat. The ratio between the electrical energy input and the heat output is the coefficient of performance (CoP). This will vary over the course of a year: the heat pump will be more efficient in the summer when the difference between the inside and outside temperatures is less.

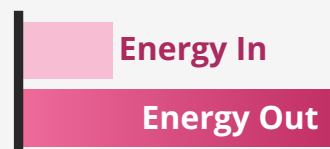
However, even in the winter the heat pump will use far less energy than a direct electrical heater or a gas boiler to produce a given amount of heat.

How efficient will my system be?

337%

From our model of your property, we expect your system to have a Seasonal Coefficient of Performance (SCoP) of 3.37.

That means for every kWh of electrical energy you put into the heat pump, you will on average get 3.37 kWh of heat.



Will I notice a difference?



Boilers typically run for short periods at very hot temperatures, and cycle rapidly on and off.

Heat pumps work more efficiently if they run at lower temperatures for longer periods, so your radiators will feel cooler than they used to but will be on for more of the time. We have based our calculations on a radiator temperature of 49.5 degrees.

You don't need to worry though - we've carefully sized your system so that you should always have enough heat to keep your house warm.

People who have heat pumps usually say they really like the more even heat that a heat pump system gives you compared with a gas boiler.

Sound check

Before your heat pump can be installed, we need to check that the noise it creates will not disturb your neighbours. Modern heat pumps are quiet, but are best not sited very close to doors or windows that may be open.

The sound check assesses how much sound from the heat pump will be transmitted to neighbouring properties. If the maximum sound level is less than 37dB then the installation can proceed without a planning application under the 'permitted development' rules.

Full details on the method used can be found in the MCS020 a) document on the MCS website.



Assessment point A

Outside

46db

No

1m between heat pump and assessment

MCS020 a) Sound Level Calculation	
General data	
Heat pump sound power level	54dB
Reflective surfaces	1
Sound pressure level	Q2

Quote

John Test
103 High View Watford Hertfordshire
WD18 6JP

Quote reference: 39
Quote date: 09/04/2026
Quote by: Tej Patel
Quote valid for: 30 days

Description of goods and services	Price
Goods	
Vaillant aroTHERM plus 3.5	
Goods total	£3,990.59
Services	
Services total	£0.00
Total Goods and Services	£3,990.59
Grant	-£7,500.00
Total before VAT	-£3,509.41
VAT 20% (goods) and 0% (services)	£798.12
Total including VAT	-£2,711.29

Order form

To proceed with this order please sign below to acknowledge that you have read and accept the information contained within this quote document and our terms and conditions.

Customer signature	Customer name	Date

Inputs

Property Details

Year built	Post 2006
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Design Data

Outside Design Temp – ODT	-2°C
Degree Days (DD)	2203
Mean air temp – MAT	11.3°C
Altitude	69m

Building Requirements

Space Heating load	2209W
Total area of building	6.25m²
Average Watts per metre square heat loss	353W/m²

Heating check

We have checked that every room in your property will still be warm enough after the installation of the heat pump.

We've surveyed every room to work out the total expected heat loss through the ventilation, roof, walls, floors, windows and doors, at the specified room temperature. We've then compared this with the heat that the heat emitters will produce to ensure that every room will be warm enough. Radiators that will be removed are displayed with a lighter shade.

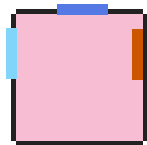
The output power of each radiator is calculated using:

The proposed flow temperature	50°C
The dT	1°C
The mean water temperature	49.5°C

Ground floor

Bedroom

Area: 6.25 m² Heat loss: 354 W/m² Design Temp: 18°C Air changes: 0.5/hr



✗ Insufficient heating 1042 W / 2210 W	
K1	As Surveyed
W1000 x H600 mm	542 W
Underfloor	New
6.25m ² , Output at 80 W/m ²	500 W

Assumptions

We have assumed an outside design temp of -2.0°C. This is based on tables of typical minimum winter temperatures in the UK.

We have used different design temperatures for different types of room. Bathrooms are typically kept warmer than living spaces such as lounges, kitchens and bedrooms for example, while utility spaces can be kept at a lower temperature. Kitchens and bathrooms also have more ventilation than other rooms, so we have allow for more heat loss through air exchange to these rooms.

Most heat loss from a property is through the building fabric however. We have provided a floorplan with a list of the materials we have used for the model along with a measure (U-value) of their insulating properties.

Heat Pump Check

Your heat pump: Vaillant aroTHERM plus 3.5

Total heat loss	2210 W
Area of building	6.25 m ²
Average heat loss	353.58 W/m ²
Output power of Vaillant aroTHERM plus 3.5 at the ODT and proposed flow temperature	4300 W



Heat pump meets demand



At the ODT -2 and proposed flow temperature 50°C, the total heat loss is less than the output power of the heat pump.

Materials

The list of materials surveyed in the building. Radiators that will be removed are displayed with a lighter shade. Floors, intermediate floors, roofs, roof glazing and dormers are not shown in the drawings but are displayed in the material list.

Ground floor



Window

 Wood Single Glazed
U-value: 4.8

.....

Door

 Glazed wood door single glazed
U-value: 4.8

.....

Radiator

 K1 - one panel, one fins

.....

External Wall

 Brick 102 mm, plaster
U-value: 2.97
Thickness: 102 mm

.....

Floor

 Solid floor with 0mm of insulation

.....

Roof

 Pitched roof - Slates or tiles, sarking felt, ventilated air space, no insulation, 9.5 mm plasterboard
U-value: 2.51

.....

MCS Performance Estimate

Under the terms of the Microgeneration Certification Scheme we are required to give you an estimate of the energy consumption of your heat pump system using a standardised procedure based on values from your Energy Performance Certificate (EPC).

Energy Performance Certificate (EPC) information

Does this estimate relate to a new build or proposal for extension or reduction in size of an existing building? Yes

EPC Number A

Energy required to heat property 12 kWh

Energy required for hot water 5 kWh

New renewable system information

Type of system Air Source Heat Pump

Manufacturer Name Vaillant

Manufacturer Model 0010037211

MCS Certification Number 12345

Flow Temperature 50°C

MCS SCOP Heating 3.37

MCS SCOP Hot water 1.75

Renewable system provides Heating and Hot Water

Hot water immersion use None

Size of hot water cylinder 5 L

Further Information

The Microgeneration Certification Scheme (MCS) is the UK's quality assurance program for certifying renewable energy products and installations.



MCS certifies both products (such as solar panels, wind turbines, and heat pumps) and the companies that install them, ensuring that products meet specific quality standards and that installations are performed by qualified and experienced professionals.

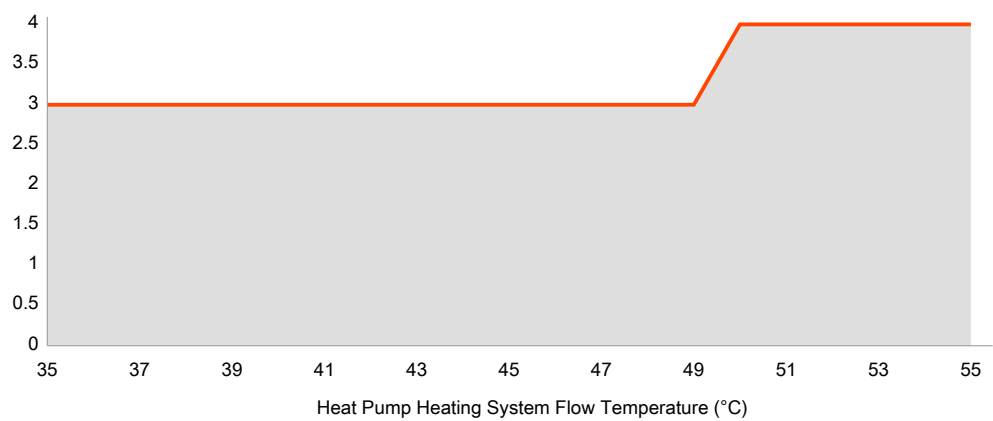
MCS also provides a mechanism for handling complaints or disputes related to certified installations.

MCS certification is a requirement for many incentive schemes, including the Boiler Upgrade Scheme grant for heat pump installations.

Estimates system performance / comparison (kWh)

	Heating	DHW	Total
Energy Requirement for the building			
Net energy required to heat property	12	5	17
Existing system consumption	24	11	35
New Heat Pump System Estimated Consumption			
Full Heat Pump System	4	3	7

Electricity Consumption of Proposed Heat Pump for Space Heating versus Flow Temperature



MCS Key facts

Predicting the heat demand of a building, and therefore the performance and running costs of heating systems, is difficult to predict with certainty due to the variables discussed here. These variables apply to all types of heating systems, although the efficiency of heat pumps is more sensitive to good system design and installation. For these reasons your estimate is given as guidance only and should not be considered as a guarantee.

Seasonal Coefficient of Performance

MCS Seasonal Coefficient of Performance (SCoP) is derived from the EU ErP labelling requirements, and is a theoretical indication of the anticipated efficiency of a heat pump over a whole year using standard (i.e. not local) climate data for 3 locations in Europe. It is used to compare the relative performance of heat pumps under fixed conditions and indicates the units of total heat energy generated (output) for each unit of electricity consumed (input). As a guide, a heat pump with a MCS SCoP of 3 indicates that 3 kWh of heat energy would be generated for every 1 kWh of electrical energy it consumes over a 'standard' annual cycle.

Energy Performance Certificate

An Energy Performance Certificate (EPC) is produced in accordance with a methodology approved by the government. As with all such calculations, it relies on the accuracy of the information input. Some of this information, such as the insulating and air tightness properties of the building may have to be assumed and this can affect the final figures significantly leading to uncertainty especially with irregular or unusual buildings.

Heat Pump SCOPs

for Vaillant aroTHERM plus 3.5

Flow Temperature	SCOP
35°C	4.41
36°C	4.34
37°C	4.26
38°C	4.18
39°C	4.11
40°C	4.03
41°C	3.95
42°C	3.88
43°C	3.80
44°C	3.72
45°C	3.65
46°C	3.59
47°C	3.54
48°C	3.48
49°C	3.43
50°C	3.37
51°C	3.32
52°C	3.27
53°C	3.21
54°C	3.16
55°C	3.10

MCS Factors & Impacts

We have identified 3 key types of factor that can affect how much energy a heating system will consume and how much energy it will deliver into a home. These are 'Fixed', 'Variable' and 'Random'. Most factors are common to ALL heating systems regardless of the type (e.g oil, gas, solid fuel, heat pump etc.) although the degree of effect varies between different types of heating system as given in the following table.

The combined effect of these factors on energy consumption and the running costs makes overall predictions difficult however an accuracy +25-30% would not be unreasonable in many instances. Under some conditions even this could be exceeded (e.g. considerable opening of windows). Therefore it is advised that when making choices based on mainly financial criteria (e.g. payback based on capital cost verses net benefits such as fuel savings and financial incentives) this variability is taken into account as it could extend paybacks well beyond the period of any incentives received, intended occupancy period, finance agreement period etc.

Factor	Impact	Energy Required
'Fixed' which include:		The heat energy output requirement of the system which directly impacts on running costs. This requirement exists regardless of the heating system chosen as it is the heat required to keep the space comfortable. Opening windows or increasing room temperatures will demand more heat output, which means more energy input but this would NOT directly affect the efficiency. Thus increased energy demand does NOT automatically mean reduced efficiency. System Efficiency The efficiency of the system has been directly affected and will therefore demand more input energy to achieve the same heat output thus increasing running costs. However, increased energy input does NOT necessarily mean lower system efficiency (see above).
Equipment Selection Performance figures (SCoP) from ErP data	System Efficiency	
Energy Assessment via the EPC (e.g. assumptions as to fabric construction and levels of insulation; the variation in knowledge and experience of Energy Assessors)	Energy Required	
'Variable' which are affected by the system design and include:		
Accuracy of sizing of heat pump- i.e. closeness of unit output selection (kW) to demand heat requirement (kW)	System Efficiency	
Design space and ambient (external) temperatures	Energy Required	
Design flow /return water temperatures, and weather compensation	System Efficiency	
Type of Heat emitter (e.g. Under-floor; natural convector (e.g. 'radiator'), fan convector etc.)	System Efficiency	
'Random' which cannot be anticipated and include:		
User behaviour:		
- Room temperature settings	Energy Required	
- Hot water usage and temperature settings	Energy Required	
- Occupancy patterns/times	Energy Required	
- Changing the design HP flow temperatures	System Efficiency	
- Ventilation (i.e. opening windows)	Energy Required	
Annual climatic variations (i.e. warmer and colder years than average)	Energy Required	